

Essential Cisco CLI Commands for Packet Tracer

Basic Navigation

- `enable` — Enter privileged EXEC mode
- `configure terminal` — Enter global configuration mode
- `exit` — Exit to previous mode
- `end` — Return to privileged EXEC mode
- `show ?` — List available “show” commands

Device Information

- `show running-config` — Show active configuration
- `show startup-config` — Show saved configuration
- `show ip interface brief` — Interface summary
- `show interfaces` — Detailed interface information
- `show version` — System/hardware information
- `show vlan brief` — VLAN table (Switches)
- `show mac address-table` — MAC learning table
- `show arp` — View ARP table
- `show protocols` — Protocol status per interface

Interface Configuration

- `interface gigabitEthernet0/0` — Access interface config
- `ip address <IP> <MASK>` — Assign IP address
- `no shutdown` — Enable interface
- `shutdown` — Disable interface
- `description <text>` — Add description to interface

Basic Device Setup

- `hostname <name>` — Set device hostname
- `banner motd #text#` — Set login banner
- `enable secret <password>` — Secure privileged mode
- `line console 0` — Enter console configuration
- `password <password>` — Set console password

- `login` — Enable console login
- `service password-encryption` — Encrypt all passwords

Testing & Troubleshooting

- `ping <address>` — Test connectivity
- `traceroute <address>` — Show network path
- `show ip route` — View routing table
- `debug ip packet` — Packet debugging (be cautious)
- `clear arp` — Clear ARP table
- `clear mac address-table` — Clear MAC table
- `show cdp neighbors` — Discover directly connected devices
- `show lldp neighbors` — LLDP neighbors (if enabled)

VLAN & Switching Commands

- `vlan <id>` — Create VLAN
- `name <name>` — Name VLAN
- `interface fastEthernet0/1` — Select switch port
- `switchport mode access` — Set access mode
- `switchport access vlan <id>` — Assign VLAN
- `switchport mode trunk` — Make trunk port
- `switchport trunk allowed vlan <list>` — Allowed VLANs

Routing Protocols

- `ip route <network> <mask> <next-hop>` — Static route
- `router rip` — Enable RIP
- `version 2` — Use RIP v2
- `network <network>` — Advertise network
- `router ospf <id>` — Enable OSPF
- `network <ip> <wildcard> area <id>` — OSPF network
- `router eigrp <asn>` — Enable EIGRP
- `network <network>` — Advertise network in EIGRP

Security Commands

- `line vty 0 4` — Configure telnet/SSH lines
- `transport input ssh` — Allow SSH only
- `username <name> secret <pass>` — Local user
- `ip ssh version 2` — Enable SSH v2
- `access-list <number> permit <conditions>` — Create ACL
- `access-class <number> in` — Apply ACL to VTY

Saving & Resetting Device

- `copy running-config startup-config` — Save configuration
- `write memory` — Save configuration (short form)
- `erase startup-config` — Remove saved config
- `reload` — Reboot the device

Useful CLI Shortcuts

- Tab — Autocomplete commands
- Ctrl + C — Cancel current command
- Ctrl + Z — Exit to privileged mode
- ? — Display context-sensitive help
- Up/Down Arrow — View command history